What the Model Says About the Twelve Steps and the Twelve Traditions

A reference primer for *Nobody in Charge*

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Revised 15 October 2026

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How to read this

This is a reference document, not a chapter. It gives one short finding for each Step and each Tradition, and it exists so that a reader who wants to know what the model says about a particular item does not have to reconstruct it from six chapters. Everything here is drawn from the manuscript and is reproducible from `model/book-calculations.ipynb`. Nothing here is new.

Every entry comes in three parts. **Technical** states the result and its provenance. **What was assumed** says, in the same plain language as the rest, which numbers in that row somebody chose rather than measured, and what would follow if the choice were wrong. **In plain terms** says what the result would mean to somebody sitting in a room, and says so without arithmetic. The third part is an interpretation of the first and is softer than it. Where the plain reading would carry further than the technical one supports, the entry says where it stops.

The middle part is there because the arithmetic in this document is exact and its inputs are not. Every figure in Part Four is a correct calculation on two tables that one person wrote down from reading, and a reader who sees only the technical line can easily mistake the precision of the calculation for the precision of the inputs. Reading the assumption beside the result is the intended use of this primer.

Three different instruments produce the findings below, and they are not interchangeable. Every entry says which one it used.

Algebra. Exact calculation on two fixed matrices. The consumption matrix S is twelve Steps by eight group resources. The governance matrix G is twelve Traditions by the same eight resources. Their product

$$B = S\,G'$$

gives the demand each Step places on resources each Tradition governs. These figures carry no sampling error. They are also no better than the two matrices, both of which one person built, and the largest outstanding item in the whole project is a second reader marking which of the ninety-six governance cells are non-zero.

Deliberation. Closed-form results from Golub and Jackson's 2010 theorem on naive learning. These concern how a room's consensus tracks the truth as the room grows. They are exact and they involve no simulation.

Simulation. Monte Carlo runs of the group model. Every figure quoted here comes from at least 400 seeds and carries an interval.

Five standing cautions

One. The coupling is not used by the simulation.

Technical. B = S G′ is a derived object. No entry of it was chosen and no entry of it feeds a run. It is a description of the two matrices, and it inherits every judgement in them.

In plain terms. The table matching Steps to Traditions was not built by watching groups. It falls out of two lists somebody wrote down: what each Step needs from other people, and what each Tradition looks after. If those two lists are wrong, everything in Part Four is wrong with them, and no amount of computing will reveal it.

Two. At full adherence the governance matrix cancels exactly.

Technical. It is column-normalised, so when every Tradition is at 1.0 the governance quality of every resource is identically 1. Thirty-five of the model's 118 registered sensitivity values cannot affect a fully adherent group at all. The count is not the total number of authored choices in the model.

In plain terms. In a group doing everything right, the Traditions do not show up in the numbers, because there is nothing left for them to fix. Everything the model says about individual Traditions is a statement about groups falling short somewhere, which is every real group. The Traditions are visible only in the breach.

Three. Sensitivity screens bound claims; they do not make them universal.

Technical. The project uses global, targeted, screening and structural designs with different estimands and power. A claim can receive strict support, tie, reverse, or remain unresolved. The corrected audit reports those categories separately and does not turn finite parameter ranges or five-seed screens into statements that a result holds for every possible model. Mortality, endpoint viability and final membership are separate outcomes.

In plain terms. Repeated tests can make a model result more or less credible inside the model, but they cannot make it a law about every specification or a forecast about AA. Read each simulation statement with its outcome, comparison, range and uncertainty. A direction that repeats is stronger than one that reverses, and an unresolved comparison is not evidence of no effect.

Four. The degradation figures rank one outcome at one point.

Technical. Where a Tradition entry quotes a membership loss, the design is: all twelve Traditions at 0.85, then one Tradition alone lowered to 0.50, twenty-year horizon, 400 paired replications under common random numbers, reference membership 13.10 with a cross-seed standard deviation of 5.67. Tradition 3 and Tradition 11 remain mixed adherence rows here; their mechanism-specific paths are separated in the release factorials.

In plain terms. That column answers one question: if a group let this Tradition slide while holding the others steady, how many fewer people would be in the room in twenty years. It is not a measure of how much a Tradition matters. It says nothing at all about whether the group's decisions are any good, which is where Traditions 2, 9 and 12 do most of their work. A

Tradition can score near zero here and still be carrying one of the book's three headline results. One of them does.

Five. Three different things were assumed, and they are not equally well tested.

Technical. Every entry below rests on three separable authored choices. First, the list of resources: the model says a group supplies eight things and no others. Second, the pattern: which of the 96 cells in each matrix are non-zero. Third, the magnitudes: how large each non-zero cell is, on a scale from 0 to 1 with no unit outside this model. The robustness designs reach these unequally. Plus or minus thirty per cent jitter and the randomized-matrix draws test magnitudes. The 64 resource-list variants test the list by deleting one resource, merging a pair, or deleting two, and they establish that the Part Four verdicts are not an artifact of any single column. They cannot test whether a resource should have been *split*, or whether a ninth resource is missing, because both require fresh judgement rather than an operation on the existing columns. Nothing tests the pattern of zeros against an outside source, which is why a second reader marking the non-zero cells is the largest outstanding item in the project.

In plain terms. There are three ways these tables could be wrong, and they are not equally guarded. The magnitudes have been shaken hard and most results survive. The list of eight has been tested by removing entries and combining them, and the headline findings hold. What has not been tested at all is the possibility that the list is missing something, or that one of the eight is really two things wearing one name. No computation can find that, because the test would have to come from outside the tables, and everything here is inside them. When a result below looks surprisingly clean, the honest question is not whether the arithmetic is right, which it is, but whether the eight columns were the right eight.

Part One. The Twelve Steps

Each entry gives the derived group-dependence coefficient beta, which is the Step's row sum in S normalised by the largest row sum, and the Step's principal supplier, which is the Tradition with the largest entry for that Step in B, the raw semantic overlap of the two authored matrices. The share-normalised map C can name a different Tradition, as it does at Step Twelve. The index-mate is the Tradition carrying the same number, and its rank among the twelve is the whole of Part Four's argument in one column.

Step One, admitting powerlessness

Technical. beta = 1.00, the joint maximum. Principal supplier Tradition 3 at 1.22, with Tradition 1 second at 0.99. Its largest consumption is identification, others in the room who name themselves as alcoholic, at the matrix maximum of 1.0, followed by admission at 0.8. It carries the highest top speed of any Step, 0.30 per week, and the lowest exposure to the maintenance gate, 0.05. Its index-mate, Tradition 1, ranks second at 0.99 against the winner's 1.22, the closest any index-mate comes. Under the sixty-four resource-list variants Step One regains its index-mate in fifteen of them, more than any other Step. *Algebra, plus the simulation's parameter table.*

What was assumed. Five of the eight resources were given non-zero entries: identification at 1.0, admission at 0.8, visible proof at 0.3, continuity at 0.2 and gentle pressure at 0.1. The 1.0 is a ceiling rather than a measurement; it says only that nothing in the table needs identification more than this Step does. Two further numbers are assumed outside the matrix: a top speed

of 0.30 per week, the fastest of the twelve, and an exposure to the maintenance gate of 0.05, the lowest of the twelve. That 0.05 is not a judgement about Step One specifically. Gate exposure was laid out as a straight line from 0.05 at Step One to 1.00 at Step Twelve, so every Step's value on that dial follows from its position rather than from anything observed about it.

The reading behind the row is that what an arrival needs is other people naming themselves the same way, and a door that opened without asking anything first. Nobody measured that. Reverse the two largest entries, so that admission outranks identification, and Tradition 3 would win this row more clearly rather than less, so the headline result here is not sensitive to that particular judgement. What the row does depend on is the claim that identification is a distinct thing a group supplies, rather than an aspect of admission. The resource-list test can delete or merge either column, and does: merging the two alone does not bring the index pairing back, but deleting both does, and Step One is among the fifteen variants where it comes back.

In plain terms. This is the Step you cannot take by yourself, and it is tied with the twelfth for depending most on other people. What it needs is not advice. It is other people in the room saying the same thing about themselves, and a door that opened without asking anything first. It is also the fastest Step to move and the one least affected by how much you already have to lose, which is the model's way of saying that an arrival has nothing yet to protect. It is the one place where the obvious pairing nearly works, and it is the Step most likely to pair with its own Tradition if somebody redrew the list of what groups supply.

Step Two, coming to believe

Technical. beta = 0.71. Principal supplier Tradition 11 at 0.82, with Tradition 5 second at 0.72. Its dominant input is visible proof that recovery happens, at 1.0. The index-mate, Tradition 2, ranks seventh at 0.13. *Algebra.*

What was assumed. Four non-zero entries: visible proof at 1.0, identification at 0.4, continuity at 0.2 and counsel at 0.1. Top speed 0.25 per week, gate exposure 0.136 from the straight line. The substantive assumption is the one large number. Coming to believe was coded as an evidential matter, so the thing it consumes most is other people visibly getting better, and it was coded as needing almost no counsel, 0.1, which is the model saying that this Step is not achieved by being talked to.

That is a reading of the Step, and a contestable one. A tradition of interpretation holds that Step Two is largely a matter of being persuaded, argued with, or taught, in which case counsel should be large and proof small. Had it been coded that way the principal supplier would move from Tradition 11 to Tradition 2, and the primer's plain-language claim that a group's best argument for itself is the condition of its members would not follow. The claim is downstream of the coding, not evidence for it.

In plain terms. The model treats coming to believe as an evidential matter rather than a persuasive one. What the Step consumes is people visibly getting better in front of you. That is why the Tradition supplying it is attraction rather than promotion: nobody is talked into this, they watch it happen and draw the obvious conclusion. If the model is right about Step Two, then a group's most important argument for itself is the condition of the people in it.

Step Three, the decision

Technical. beta = 0.29, near the bottom. Principal supplier Tradition 2 at 0.31, with Tradition 1 second at 0.27. Row sum 0.70, so it asks little of the group in absolute terms. Index-mate

Tradition 3 ranks seventh at 0.01, the smallest non-zero index-mate entry in the matrix. *Algebra.*

What was assumed. Four small non-zero entries: counsel at 0.3, visible proof at 0.2, continuity at 0.1 and gentle pressure at 0.1, giving the third smallest row sum in the table at 0.70. Top speed 0.25 per week, gate exposure 0.223. Every number here is small, and that smallness is itself the assumption: the decision was coded as something a person does, with a group nearby rather than involved.

This matters more than the individual cells, because the starkest number against the index-pairing idea anywhere in the matrix, Tradition 3 ranking seventh at 0.01, is partly a consequence of the row being small in total. A row that asks little of a group gives every Tradition little, and rank is then decided by fine differences. The finding survives the magnitude jitter, so it is not fragile in that sense. But a reader should know that the dramatic phrasing rests on a row the author deliberately made quiet.

In plain terms. The decision asks less of a group than almost any other Step. What it does draw on is counsel: somebody to talk it over with. The Tradition sharing its number contributes essentially nothing to it, and that is the starkest non-zero number against the pairing idea anywhere in the matrix.

Step Four, the inventory

Technical. beta = 0.33. Principal suppliers, tied: Traditions 1 and 2, both exactly 0.35 in the raw overlap B (the normalised map C separates them in favour of Tradition 1). The two are equal in exact arithmetic, so Step Four has no single principal supplier in B, and this is the one exact tie at the top of any row. Its largest consumption is gentle pressure at 0.4 and counsel at 0.3. The index-mate, Tradition 4, has an entry of exactly zero, because Tradition 4 governs no resource any Step consumes. That zero is arithmetic and not evidence: no multiplicative perturbation can move it. The threshold test that can reach it finds that a uniform governance strength of 0.438 would be needed for the index-mate to win, against a mean live entry in the governance matrix of 0.374. *Algebra.*

What was assumed. Three non-zero entries: gentle pressure at 0.4, counsel at 0.3 and identification at 0.1. Top speed 0.18 per week, among the slowest, and gate exposure 0.309. The reading is that an inventory needs mild expectation that you will actually do it and somebody to ask about how, and needs nothing else from a group.

The index-mate's exact zero is not an assumption about Step Four at all. It follows from a decision made in the other table, that Tradition 4 governs no resource any Step consumes. No perturbation of Step Four's own row can move it, because zero times anything is zero. This is the clearest case in the primer of a result that looks like a finding and is really a restatement of an input, which is why the entry says so rather than counting it.

In plain terms. The inventory needs two things from a group: mild expectation that you will actually do it, and somebody to ask about how. Tradition 4 supplies neither, and it supplies nothing to any other Step either, because it is one of the five Traditions in the model that guard rather than provide. So the pairing fails here for a boring reason rather than an interesting one, and the primer says so rather than counting it as a discovery.

Step Five, telling someone

Technical. beta = 0.71. Principal supplier Tradition 12 at 1.08, with Tradition 1 second at 0.65, a margin of 0.43, the second widest in the matrix. Step Five is the only Step that consumes confidentiality at the matrix maximum of 1.0, and Tradition 12 is the Tradition that governs confidentiality at 1.0. Its robustness is asymmetric: it survives 99.5 per cent of draws at plus or minus thirty per cent jitter, with a 95 per cent Wilson interval of [99.1, 99.7] on 2,000 draws, and only 17.5 per cent [15.9, 19.2] when every non-zero magnitude is replaced at random. Its index-mate, Tradition 5, ranks fifth at 0.12. *Algebra.*

What was assumed. Five non-zero entries, one of them decisive: confidentiality at 1.0, counsel at 0.3, continuity at 0.2, identification at 0.1 and gentle pressure at 0.1. Top speed 0.22 per week, gate exposure 0.395. Two separate judgements produce the cleanest match in the book, and they were made in two different tables: that Step Five is the only Step needing confidentiality at full strength, and that Tradition 12 supplies confidentiality at full strength.

The entry's own robustness numbers are the honest measure of that. Holding the pattern of which cells are non-zero and jittering the magnitudes by thirty per cent, the match survives 99.5 per cent of the time. Replacing every non-zero magnitude at random, so that only the pattern remains, it survives 17.5 per cent of the time. The gap between those two figures is not noise. It is the size of the authored judgement, stated numerically, and it is the reason this entry is described as a judgement argued for rather than a result computed.

In plain terms. Telling someone requires that it stay told. Step Five is the only Step that needs confidentiality at full strength, and anonymity is the only Tradition that supplies it at full strength. That is the cleanest match in the book, and it is also the clearest illustration of what these matches rest on. If you accept my numbers it is near certain. If you accept only the pattern of which Tradition touches what, and let the strengths fall where they may, it mostly disappears. The pairing is a judgement argued for, not a result computed, and the difference between ninety-nine and seventeen is exactly the size of the judgement.

Step Six, becoming willing

Technical. beta = 0.25. Principal supplier Tradition 1 at 0.25, with Tradition 2 second at 0.24, a margin of 0.01 and the narrowest of any row except Step Four's, which is a tie. Row sum 0.60. Index-mate Tradition 6 is a structural zero and ranks seventh, tied with the other zeros in its row. Its threshold is 0.417, the lowest of all twelve. *Algebra.*

What was assumed. Three non-zero entries: gentle pressure at 0.3, counsel at 0.2 and visible proof at 0.1, giving a row sum of 0.60. Top speed 0.20 per week, gate exposure 0.482. As with Step Three the assumption is mostly the smallness.

The technical line reports that the top two Traditions are separated by 0.01, which is below any precision the inputs can support. That is worth stating plainly: where the margin between two Traditions is smaller than the rounding in the table that produced it, the primer should be read as saying the model has no opinion, and not as saying Tradition 1 narrowly wins. The entry is phrased that way deliberately.

In plain terms. Becoming willing barely needs a group at all: a little pressure, a little counsel, and that is most of it. The two Traditions at the top are so close that the model has no real opinion about which one carries it. By the threshold test, of all twelve Steps this is the one where the pairing comes nearest to being recoverable, and it still needs its own Tradition to look after its

needs more attentively than a typical entry in the whole matrix.

Step Seven, asking

Technical. beta = 0.17, the minimum. Row sum 0.40, the smallest in S. Principal supplier Tradition 1 at 0.17. The ratio of the most group-dependent Step to this one is 6.0. Index-mate Tradition 7 is a structural zero and ranks seventh, tied with the other zeros in its row. Setting a uniform governance strength at the mean live entry of 0.374 for all twelve Steps at once, Step Seven's index-mate loses by the narrowest margin of the twelve, 0.02. *Algebra.*

What was assumed. Three non-zero entries: gentle pressure at 0.2, visible proof at 0.1 and counsel at 0.1, giving the smallest row sum in the whole table at 0.40. Top speed 0.20 per week, gate exposure 0.568.

The six-to-one ratio between the most group-dependent Step and this one is therefore an assumption's shadow rather than a discovery. It is arithmetic on a row the author made the smallest, because the reading was that asking is the most private thing in the programme. The ratio is worth quoting because it makes the reading explicit and falsifiable: anyone who thinks Step Seven needs more from a room than Step Four does has a specific disagreement with a specific number, which is the most this kind of table can offer.

In plain terms. This is the most private Step in the programme. If you asked how much of each Step happens in a room rather than in a person, Step Seven gives the smallest answer, by a factor of six against the largest. That is not a claim that it is easy, or minor, or that it happens without the rest. It is a claim about how much of it other people can supply, and the answer the model gives is: almost none of it.

Step Eight, listing the harms

Technical. beta = 0.33. Principal supplier Tradition 2 at 0.43, with Tradition 1 second at 0.29. Its largest consumption is counsel at 0.4. The index-mate, Tradition 8, ranks fourth at 0.07, level with Step Eleven's, the best rank achieved by any index-mate other than Step One's. *Algebra.*

What was assumed. Three non-zero entries: counsel at 0.4, gentle pressure at 0.3 and confidentiality at 0.1. Top speed 0.18 per week, gate exposure 0.655. The reading is that making the list is mostly a matter of having somebody to ask, with some expectation that it gets done and a little need for discretion.

The single interesting assumption is the small confidentiality entry. It gives Tradition 8 its 0.07 here, which is not its largest entry in the table (that is 0.34, at Step Five). Remove that 0.1 and the entry falls to 0.04, tied with Tradition 12, though its competition rank stays fourth. The rank therefore does not rest on that tenth of a unit, but the 0.07 does. Its fourth place, shared with Step Eleven's, is the best index-mate rank in the table after Step One's second.

In plain terms. Making the list is mostly a matter of having somebody to ask, which is why the group conscience supplies it. Its own numbered Tradition does better here than in most rows and still comes fourth, which is the pattern across the whole table: the pairing is not absurd anywhere, and it wins nowhere.

Step Nine, amends

Technical. beta = 0.63. Principal supplier Tradition 2 at 0.90, with Tradition 1 at 0.48, a margin of 0.42 and the third widest in the matrix behind Step Ten's 0.51 and Step Five's 0.43. Step Nine

consumes counsel at 0.9, the largest entry in S other than the four entries at 1.0. It carries the lowest top speed of any Step, 0.15 per week. Disabling the twelfth Step lowers mean Step Nine practice by 0.0055 in the 400-seed paired experiment, but the 95 per cent interval is -0.0015 to 0.0125 and includes zero. The same intervention does produce resolved decreases in membership, established practice and maintenance capacity. Index-mate Tradition 9 is a structural zero and ranks eighth, tied with the other zeros in its row. *Algebra and simulation.*

What was assumed. Four non-zero entries: counsel at 0.9, confidentiality at 0.3, gentle pressure at 0.2 and continuity at 0.1. The 0.9 is the largest entry in the table that is not one of the four maxima. Top speed 0.15 per week, the slowest of the twelve, and gate exposure 0.741.

Two of those numbers were chosen to say something the author believed and did not measure: that amends is the Step most needing somebody to talk it through with, and the one that moves slowest. Both are readings of the practice rather than observations of it, and the plain-language sentence about not doing this one quickly is a restatement of the assumed speed, not a finding about it. The finding in this entry is the separate simulation result about switching off the twelfth Step, which does not depend on Step Nine's own row.

In plain terms. Amends is the Step that most needs counsel, and it is the slowest Step to move, which between them describe something people already know: you do not do this one quickly and you do not do it without asking. The finding worth carrying is not about Step Nine's own needs. Switch off the twelfth Step, so nobody in the group is carrying the message to anyone, and the group ends smaller with lower maintenance capacity. The isolated Step Nine change is too imprecise to call. The model therefore supports a group-level service pathway here, not the earlier claim of a large resolved spillover into Step Nine itself.

Step Ten, the daily inventory

Technical. beta = 0.63. Principal supplier Tradition 1 at 0.94, with Tradition 2 second at 0.43, a margin of 0.51, the widest in the matrix. Its largest consumptions are gentle pressure at 0.7 and week-to-week continuity at 0.5, which between them are the two resources Tradition 1 supplies most heavily. Index-mate Tradition 10 is a structural zero and ranks eighth, tied with the other zeros in its row, which is the worst rank in the table, shared with Steps Four and Nine. *Algebra.*

What was assumed. Four non-zero entries: gentle pressure at 0.7, continuity at 0.5, counsel at 0.2 and confidentiality at 0.1. Top speed 0.25 per week, gate exposure 0.827. The reading is that a daily inventory is a habit, and that habits are held in place by the meeting happening again and by other people mildly expecting you at it.

The widest margin in the matrix, 0.51 in Tradition 1's favour, is produced by that reading meeting a matching one in the other table, where Tradition 1 was given the two largest shares of exactly those two resources: 60 per cent of gentle pressure and 50 per cent of continuity. The two tables were written by the same person, so agreement between them is not independent corroboration. This row is the clearest place in the primer where a striking number comes from one judgement appearing twice.

In plain terms. Daily inventory is a habit, and habits need the two things a group supplies steadily rather than dramatically: the meeting happening again next week, and other people mildly expecting you at it. That is the whole of what the model means by unity here, and it is a duller thing than the word suggests. Step Ten also has a joint-worst showing for the pairing idea. Its own numbered Tradition ties for the worst rank in the table.

Step Eleven, the conscious contact

Technical. beta = 0.33. Principal supplier Tradition 1 at 0.47, with Tradition 2 second at 0.21. Index-mate Tradition 11 ranks fourth at 0.11. With Steps Ten and Twelve it forms the maintenance capacity term: the model computes a member's capacity to hold what they have as a Hill function of the mean of Steps Ten, Eleven and Twelve, and that capacity multiplies the growth of every Step, weighted by an exposure rising from 0.05 at Step One to 1.00 at Step Twelve. *Algebra and the simulation's structure.*

What was assumed. Four non-zero entries: gentle pressure at 0.4, continuity at 0.2, visible proof at 0.1 and counsel at 0.1. Top speed 0.20 per week, gate exposure 0.914. The row itself is unremarkable and was meant to be.

The consequential assumption about this Step is structural and lives outside its row. The model computes maintenance capacity from the mean of Steps Ten, Eleven and Twelve, and that grouping was chosen rather than derived: three Steps were nominated as the maintenance Steps and the other nine were not. The straight-line gate exposure then decides how much that capacity matters to each Step, from 0.05 at Step One to 1.00 at Step Twelve. Both the choice of which three, and the straightness of that line, are modelling conveniences. The plain-language claim that the last three Steps stop the first nine leaking away is a description of that construction, and would be false in a model that nominated a different three.

In plain terms. Its own row is unremarkable. Its importance is structural: along with Ten and Twelve it is what the model calls maintenance, the capacity to keep hold of what has already been gained. Those three set a multiplier on everything else, weighted so that it barely touches Step One and fully governs Step Twelve. In plainer language, the last three Steps are what stop the first nine leaking away, and the more practice a modeled member has accumulated, the more there is to maintain. The model records no tenure, so this state comparison cannot be translated into newcomer and veteran cohorts.

Step Twelve, carrying it

Technical. beta = 1.00, the joint maximum, and the Step that consumes seven of the eight resources. Principal supplier Tradition 5 at 1.25, with Tradition 3 second at 1.10. It is the only consumer of the recipient resource, which is the only one of the eight defined by a ratio of member states: low-practice members per high-practice potential helper. It records no tenure, sponsorship or matching, and lower helper count increases rather than reduces the ratio. Three simulation results at 400 paired seeds, thirty years, full adherence. Setting the twelfth Step's growth rate to zero lowers endpoint membership from 17.80 plus or minus 0.88 to 12.48 plus or minus 0.35; the paired loss is 5.33 [4.44, 6.21]. Endpoint viability is 394 of 400 in both conditions. In the corrected clean ablation, forcing only recipient capacity to one raises final membership by 1.03 members with paired 95 per cent interval [-0.26, 2.31], so the effect is unresolved. Index-mate Tradition 12 ranks sixth at 0.12, and this is the fragile row of Part Four: the top two are 1.25 and 1.10, a margin of 0.15, and the Step Twelve to Tradition 5 assignment survives only 67.4 per cent [65.3, 69.4] of draws at plus or minus thirty per cent jitter. *Algebra and simulation.*

What was assumed. Seven of the eight resources are non-zero, the most of any Step: recipient opportunity at 1.0, continuity at 0.6, gentle pressure at 0.3, admission at 0.2, and identification, visible proof and counsel at 0.1 each. Top speed 0.22 per week, gate exposure 1.00, the maximum, again by position on the straight line.

The recipient resource carries the weight here and is the most heavily assumed object in the

model. It is the only one of the eight defined as a ratio of member states, low-practice members per high-practice potential helper, and that definition has a consequence the author did not choose but must own: a group with fewer experienced members has a *higher* ratio and therefore looks richer in this resource, not poorer. It records no tenure, no sponsorship, and no matching between a particular helper and a particular newcomer. Whether that ratio is a reasonable stand-in for the opportunity to be useful is a hypothesis. The thirty per cent membership loss is not shown to depend on it: forcing recipient capacity alone changes membership by 1.03 [-0.26, 2.31], which is unresolved, while disabling Step Twelve also removes its attraction and maintenance roles. The row is also the fragile one in Part Four for a separate reason: the top two Traditions are 1.25 and 1.10, and the assignment to Tradition 5 survives only 67.4 per cent of magnitude draws.

In plain terms. Carrying the message is not a reward collected at the end inside this model. Turn its growth off and the group loses about thirty per cent of mean endpoint membership, while the estimated viability fraction is unchanged. The experiment ends at thirty modeled years and does not establish that either condition stays there indefinitely. The recipient calculation is a hypothesis about opportunity per potential helper, not evidence about how real sponsorship matches form. The clean comparison is too imprecise to say that removing the modeled limit improves the group. The earlier stronger conclusion came from an intervention that also changed Step weights and beta.

What the Step rows say taken together

Technical. Group-dependence is highest at the two ends, 1.00 at Step One and Step Twelve, and lowest in the middle, 0.17 at Step Seven, with a mean of 0.53. That the entry Step and the service Step depend most on other people is a consequence of the resource assignment rather than an input to it. Beta is a within-model dependence index that blends autonomous and resource-supported peer growth. The code does not implement a reciprocal member-to-group weight, so no one-to-six transmission ratio follows. The ordering claim, that a Step cannot be skipped, is a limiting case: writing the stage as a constant-elasticity-of-substitution aggregator, strict ordering holds for every substitution parameter rho less than or equal to zero. The current simulation is only a proxy-averaging demonstration: it supplies the regressors and loadings without error and adds noise to the output. More output proxies improve resolution in that exercise, but it does not validate a latent-variable estimator or a required instrument count.

In plain terms. The Steps form a U. The first and the last are things you cannot do alone. The middle ones are largely yours to do, with a group nearby rather than involved. Nobody designed that shape and it was not put in by hand; it comes out of asking, for each Step in turn, what it needs from other people.

The second point is about the rule that you cannot skip a Step. That has always been stated as a piece of folk wisdom, take it or leave it. The model shows it is not a separate belief at all. It is one setting of a dial that runs continuously from "each Step strictly requires the one before" to "the Steps substitute freely for each other", and the folk rule is everything on one half of that dial rather than a single extreme point. Which means it is the kind of claim that could be measured. The calculation here makes the estimand explicit and shows that averaging noisy outputs can improve precision under oracle information. It does not establish a sufficient sample size or measurement design for real data, where inputs are latent and group attention is endogenous.

Part Two. The Twelve Traditions

Each entry gives the Tradition's load, which is its column sum in B and therefore the total demand all twelve Steps place on resources it governs, how many of the eight resources it governs, and its membership loss in the degradation comparison described in caution four. Losses whose 95 per cent interval includes zero are marked unresolved: seven of the twelve comparisons resolve and five do not. The T3 and T11 entries instead lead with the more informative 400-seed path-split factorials.

Tradition 1, unity

Technical. Load 6.52, the largest, governing all eight resources and the only Tradition to do so. Principal supplier for four Steps, runner-up for five, and joint first with group conscience at Step Four, so it is in the top two for ten of the twelve. Degrading it alone costs 1.65 members [0.99, 2.30], third largest. Its primacy survives 75.4 per cent [73.5, 77.2] of fully structural draws, against 40.6 per cent for the rejection of index-pairing. It is not the most diffuse Tradition by concentration: singleness of purpose scores marginally lower at 0.197 against unity's 0.204. Its load is carried by two resources: continuity at 1.89 and gentle pressure at 1.86 are 57.5 per cent of the total, and transferring either one to Tradition 5 makes Tradition 5 the leader. Six of eight such transfers cannot flip it and two can. *Algebra, with simulation for the degradation figure.*

What was assumed. Unity is the only Tradition given a non-zero entry in all eight columns: continuity 0.9, gentle pressure 0.6, identification 0.5, visible proof 0.3, confidentiality 0.3, admission 0.2, counsel 0.2 and recipient opportunity 0.1. Because the governance table is column-normalised, what those numbers do is set shares. Unity ends up owning 60 per cent of gentle pressure and 50 per cent of continuity, and much smaller shares of everything else.

The decision to let one Tradition touch every resource is the single most consequential choice in the governance table, and it is what makes unity the largest load by a wide margin. It was not forced. A reader who thinks unity is a property that emerges from the other eleven rather than a supplier alongside them would give it an empty row, as the five protective Traditions have, and unity's primacy would vanish by construction rather than by argument. The project's answer is not that the choice is obviously right but that it is testable in one specific place: unity's lead depends on holding continuity and gentle pressure, and moving either to Tradition 5 hands over the lead.

In plain terms. Unity is not one of the things a group supplies. It is the condition of everything a group supplies, which is why it is the only Tradition touching all eight and why it is in the top two for ten of the twelve Steps. But look at what that actually consists of, and most of it is two unglamorous things: the meeting keeps happening, and people notice whether you are at it. Take those two away and unity drops to third place. So the model's "unity" is closer to reliability and mild social expectation than to fellow feeling, and a group worried about its unity would do better to check whether it has cancelled a meeting than to check whether everyone is getting on.

Tradition 2, the group conscience

Technical. Load 3.89, second largest, governing four resources, with counsel at 0.69 of its load and the highest concentration of any Tradition at 0.520. Principal supplier for three Steps, runner-up for four, and joint first with unity at Step Four. Degrading it alone costs 0.48 members [-0.14, 1.10], an unresolved contrast. In the deliberation model it is one of three Traditions keeping maximum influence falling toward one over N, which is the Golub and Jackson condition for a consensus converging on the truth. A single member holding 0.35 of every row floors the group's error at 0.279 however large the group grows, against a flat error falling as one over the square root of N. Five per cent of every row leaves the error 1.84 times the flat benchmark at a thousand members, and the factor grows without bound. *Algebra, deliberation and simulation.*

What was assumed. Four non-zero entries: counsel 0.9, continuity 0.2, gentle pressure 0.2 and confidentiality 0.1. After column normalisation that makes the group conscience the owner of 69 per cent of all counsel in the model, the second most concentrated position any Tradition holds over any resource, after Tradition 3's 71 per cent of admission.

That single assumption is doing nearly all the work in this row, and it encodes a specific reading: that when somebody in a meeting has somebody to ask, what they are drawing on is the group's collective judgement rather than an individual friendship. It is a defensible reading and it is not the only one. Note also that the row and the theorem are independent of each other. The Golub and Jackson result about a single member holding influence does not come from this table at all, and would stand unchanged if every entry in this row were different. That is why the deliberation findings are the most robust things in the primer and the load figure is among the least.

In plain terms. In the simulation, the group conscience is mostly what supplies counsel: it is the Tradition behind there being somebody to ask. Its real work is somewhere the membership numbers cannot see it. It is what keeps a room deciding by adding up what everybody thinks rather than by deferring to one person, and the theorem behind that is unforgiving. A group that leans on one member is permanently worse at being right, and it does not matter how big the group gets, because the leaning does not dilute. Nor does the room have to be dominated for this to bite. A member who holds five per cent of everyone's attention, which is not much and would not look like a problem from inside, nearly doubles how wrong a large group ends up. The damage starts long before anybody would call it a problem.

The objection AA itself raises. The fellowship's 1953 commentary on this Tradition describes a group's committee as sharply limited, unable in any sense to govern or direct, and then says where the influence actually is: with elder statesmen, former officeholders it calls the real and permanent leadership, who become the voice of the group conscience and to whom a perplexed group inevitably turns. They hold no office, so nothing rotates them out. Three such members holding a tenth of the attention between them cost a group of ten almost nothing, one per cent, and cost a group of a thousand a factor of 2.08 on error, with their share settling at 0.034 against a flat benchmark of 0.001. Widening the rotation does not touch it, because the concentration is not in the rotation. This is a claim about a described practice and not about the Tradition, which still satisfies the condition; but a group following AA's own commentary faithfully will build the thing the condition forbids. *Deliberation model.*

Tradition 3, the open door

Technical. Raw semantic load 2.69, governing four resources, with admission at 0.37 of that raw load. In the 400-seed paired factorial, the baseline ends at mean N 17.80 [16.92, 18.68]. Friction loss alone costs 2.96 members [1.85, 4.08]; governance loss alone costs 6.03 [5.04, 7.01]; combined loss costs 11.05 [10.03, 12.06]. The interaction is -2.06 [-3.41, -0.70], so the two contrasts must not be added. Combined loss closes 25.0 per cent of groups [21.0, 29.5] and leaves 54.8 per cent endpoint-viable [49.9, 59.6]. Because the Tradition removes the group's power to refuse admission, Tradition 3 does not appear in the default arrival rate. It has two other paths: it governs four resource columns and it reduces an inverse-practice-weighted dropout friction. The latter is largest for members whose practice is near zero; it measures neither tenure nor demographic newness. The corrected factorial reports governance loss, friction loss, their combination and their interaction separately. Older combined sweeps cannot be described as retention only. *Algebra and simulation.*

What was assumed. Four non-zero governance entries: admission 1.0, recipient opportunity 0.8, identification 0.4 and continuity 0.1, giving it 71 per cent of admission and a third of recipient opportunity. Two further assumptions matter more than the cells. The first is an absence: because the Tradition removes a group's power to refuse anyone, Tradition 3 was deliberately left out of the arrival rate, so in the default model an open door does not bring more people through it. The second is the dropout friction path, where the effect is weighted by the exponential of minus six times a member's mean practice.

That weighting is an assumption with a specific and easily misread consequence. It concentrates the effect on members whose practice is near zero, and the model has no concept of tenure, so those are not newcomers. They are members with little practice, who may have been in the room for years. Every plain-language reading of this Tradition has to be policed on that point, and the earlier version of this primer failed to.

In plain terms. In the default model a group cannot decide who turns up, but it can affect what people receive and whether low-practice members stay. Those are two distinct mechanisms, and the old description collapsed them into one. The model records practice, not arrival date, so a room's low-practice fraction must not be translated into a count of newcomers. Testing demographic newness would require tenure or cohort data the model does not have.

Tradition 4, autonomy

Technical. Load 0.00. Its row in the governance matrix is identically zero: it governs no resource any Step consumes. It is one of the five protective Traditions and enters the simulation only as a multiplier, paired with Tradition 7, on the effective adherence of everything else. Degrading it alone costs 0.88 members [0.26, 1.50], identical to Tradition 7's to the last digit because the two enter the same term symmetrically. That identity is an artefact of the model's construction, not a finding about the Traditions. *Algebra and simulation.*

What was assumed. The empty row is the assumption. Autonomy was coded as governing none of the eight resources, which makes its load exactly zero, and that zero then propagates into every Step's index-mate comparison as an unbeatable disadvantage. Its only route into the simulation is a multiplier it shares with Tradition 7.

Both halves of that were chosen. Deciding that a Tradition protects rather than supplies is a judgement about what kind of thing it is, and pairing it with self-support in a single symmetric term is a modelling convenience with no evidential content whatever. The entry already says

the identical membership figures are an artefact; the deeper point is that the whole protective category is an authored partition of the twelve, not a result. Five Traditions were placed on one side of it and seven on the other before any calculation was done.

In plain terms. Autonomy hands the Steps nothing directly. Its whole job in the model is keeping the other Traditions from being overridden from outside, and it does that jointly with self-support. The two share a number because the model treats them as a pair, not because anybody discovered they were equally important. What *is* derived here is only that autonomy supplies nothing; how it should act instead was assumed, and the multiplier is the simplest assumption that gives it any role at all.

Tradition 5, singleness of purpose

Technical. Load 3.88, a hair behind the group conscience, governing six resources, the second broadest. Principal supplier for Step Twelve and runner-up for Step Two. Degrading it alone costs 0.96 members [0.35, 1.57]. Transferring unity's governance of continuity or of gentle pressure to singleness of purpose makes it the leader, which is the strongest form of the objection that later AA literature substituted unity for single-purposedness. Six of the eight possible transfers cannot flip it and two can. *Algebra and simulation.*

What was assumed. Six non-zero entries: recipient opportunity 0.9, visible proof 0.5, identification 0.4, continuity 0.3, gentle pressure 0.2 and admission 0.1. Its share of recipient opportunity, 38 per cent, is the largest any Tradition holds of that column, which is what makes it the principal supplier of Step Twelve.

The contest with unity described in the technical line is therefore a contest between two authored rows, and the numbers deciding it are 0.9 and 0.6 in one row against 0.3 and 0.2 in the other on the same two resources. Nothing outside the author's reading placed them that way. This is the most honest place in the primer to see what the whole apparatus rests on. A real and unsettled question about AA, whether unity or single-purposedness is the load-bearing idea, is here decided by four numbers one person wrote down. The project's contribution is not the answer but the demonstration that those four numbers are where the answer lives.

In plain terms. Singleness of purpose is unity's only real rival for the most load-bearing Tradition, and the whole contest comes down to who owns two things: the meeting continuing to happen, and the pull of other people expecting you there. If those belong to unity, unity leads. If they belong to the group having one job, single purpose leads. That is a real question about how AA works and the model cannot settle it. What the model can do is say that this, and nothing else in the matrix, is where the answer would change.

Tradition 6, no endorsement

Technical. Load 0.00, a protective Tradition with an empty governance row. Enters the simulation as a multiplier, paired with Tradition 10, on the effective adherence of singleness of purpose. Degrading it alone costs 0.23 members [-0.24, 0.69], **unresolved**: the comparison cannot distinguish its effect from zero at 400 paired replications. Identical to Tradition 10's figure for the same structural reason as the Tradition 4 and 7 pair. *Algebra and simulation.*

What was assumed. An empty governance row, and a multiplier shared with Tradition 10 acting on the effective adherence of singleness of purpose. The pairing is the interesting assumption: it says that refusing endorsements and refusing outside issues protect the same thing, and protect it in the same way, so the model gives them one mechanism between them.

That is a claim, not an observation, and it has the same consequence as the Tradition 4 and 7 pairing: the two members of the pair are guaranteed identical numbers before any simulation runs. When two Traditions in this primer report the same figure to the last digit, that is always construction and never evidence.

In plain terms. Refusing endorsements supplies nothing to any Step. Its job is protecting the group's single purpose from being diluted, alongside Tradition 10. The simulation cannot tell its effect apart from nothing at all, and the correct reading of that is that the simulation has nothing to say about it, not that it does nothing. An instrument that cannot resolve a thing is silent about it, not against it.

Tradition 7, self-support

Technical. Load 0.00, protective, empty row. Paired with Tradition 4 as a multiplier guarding against outside override. Degrading it alone costs 0.88 members [0.26, 1.50], identical to Tradition 4's by construction. Money enters the simulation elsewhere, through a solvency term scaling the continuity resource: a group whose established members fall short of the rent supplies continuity somewhere between 1.0 and a floor of 0.45, nearer the floor the fewer established members it has. That mechanism is not attributed to Tradition 7 in the governance matrix. *Algebra and simulation.*

What was assumed. An empty governance row, and the shared multiplier with Tradition 4. The consequential assumption is where money went instead. The model does represent solvency, through a term that scales the continuity resource down from 1.0 toward a floor of 0.45 as a group's established members fall short of what it takes to cover the rent, but that term was attached to the group's finances directly rather than to the Tradition that produces them.

So the zero in this row is not a claim that self-support does not matter. It is a bookkeeping decision about where to attach an effect the model plainly contains. The entry says a second person building the table would likely disagree here, and this is what they would be disagreeing with: not a finding, but an attribution.

In plain terms. Passing the basket looks after nothing directly in the matrix, and yet money is unmistakably in the model somewhere else: a group that cannot pay its rent supplies week-to-week continuity at well below full strength, and continuity is one of the two things holding unity up. So the practical content of self-support is that the meeting keeps happening. The matrix does not give Tradition 7 credit for that, and a reader could reasonably say it should. This is one of the places a second person building the governance table would most likely disagree with the first.

Tradition 8, non-professional

Technical. Load 1.11, the smallest non-zero load, governing four resources with confidentiality the largest at 0.41 of the total. Principal supplier for no Step and runner-up for none. Degrading it alone changes membership by -0.20 members [-0.71, 0.31], **unresolved**. *Algebra and simulation.*

What was assumed. Four small non-zero entries: confidentiality 0.3, and identification, visible proof and counsel at 0.1 each, the smallest set of non-zero values given to any Tradition that has any. The reading is that keeping the fellowship unpaid mainly protects candour, which is why confidentiality is the largest of the four.

Everything the primer says about this Tradition follows from those four small numbers, which

is why the entry says the model can say least about it. The unresolved membership figure is not independent evidence of unimportance: a row built small will produce small effects, and the design cannot separate a Tradition that does little from one the author coded as doing little. Both readings fit the same output.

In plain terms. This is the Tradition the model can say least about. It touches four things lightly, leads on none of them, and its cost in members cannot be told apart from zero. Stating that plainly is better than dressing it up. Keeping AA unpaid may matter a great deal for reasons this model was never built to see, and the honest report is that the instrument did not detect anything rather than that there is nothing there.

Tradition 9, no hierarchy

Technical. Load 0.00, protective, empty row. Degrading it alone costs 0.02 members [-0.41, 0.46], **unresolved**. In the deliberation model, rotation of service works only if the rotating pool scales with the group. A fixed pool floors maximum influence at roughly the officeholder's share divided by the pool size, while the flat benchmark keeps falling as one over N, so the gap grows without limit: a pool of twelve sits at 2.1 times the flat benchmark at fifty members and 23.9 times at eight hundred. The pool needed to stay within a factor of two of flat is 26 per cent of the group at every size tested from fifty to eight hundred. The 26 per cent depends on the parameter choices; the divergence does not. *Algebra, deliberation and simulation.*

What was assumed. An empty governance row, so the entire membership column result for this Tradition is a consequence of a coding decision rather than a measurement.

The rotation finding is the important thing here and it rests on different assumptions altogether. It comes from the deliberation model, not from the matrices, and it assumes that influence is concentrated in whoever currently holds a service position and is shared evenly within the rotating pool. That is a stylised picture of how attention works in a room. What makes the result durable is that its conclusion does not depend on the details: any fixed pool in a growing group floors influence above the flat benchmark, so the divergence follows from the arithmetic of a constant divided by a growing number. The 26 per cent is parameter-dependent and is offered as an order of magnitude. The divergence is not, and that distinction is why this entry carries a headline result while its membership figure carries nothing.

In plain terms. This is the sharpest warning against reading the membership column as importance. On that column Tradition 9 does nothing measurable, and it carries one of the three headline results in the book.

Rotation is where the model knows something AA does not say, and since 17 August 2026 that can be put more precisely. The Twelve Concepts for World Service, adopted in 1962 and the fellowship's fullest statement on service structure, do contain a proportionality principle: Concept 4 asks for voting representation in reasonable proportion to the responsibility each part of the structure discharges. But it proportions voting weight to responsibility, not the rotating pool to the size of the group. A structure could satisfy Concept 4 exactly and still rotate twelve people through a fellowship of eight hundred. AA has the instinct for proportion and applies it to representation rather than to the pool.

The long form of this Tradition says rotating leadership is best, and the 1953 commentary warns against entrenched power, but neither says how many people. The model says the answer is a fraction of the group and not a headcount, and that the difference is not a matter of degree. A group of fifty rotating twelve people through its service positions is fine. A group of eight hun-

dred rotating twelve is a permanent oligarchy, whoever those twelve are and whatever anybody intends, and from inside it looks exactly like the healthy small group did. Roughly a quarter of the group needs to be in the pool. Treat the quarter as an order of magnitude rather than a target, and treat the underlying question as the durable one: if this group doubled, would the same people still be running it?

And a limit on that advice, from AA's own commentary. The quarter is necessary and it is not sufficient. The same 1953 text that recommends rotation says the rotating positions carry no governing authority, and locates real influence in elder statesmen who hold no position at all. A group rotating a quarter of itself but deferring to three such members sits at 2.5 times the flat benchmark at fifty, 9.2 at two hundred and fifty, and 34.2 at a thousand. A service roster cannot detect this, because a roster records offices and nothing records deference. The second question a group would have to ask is: when something difficult comes up, how many different people does this room turn to, and is that number growing as the room does?

Tradition 10, no outside issues

Technical. Load 0.00, protective, empty row. Paired with Tradition 6 as a multiplier on singleness of purpose. Degrading it alone costs 0.23 members [-0.24, 0.69], **unresolved**, identical to Tradition 6's by construction. Part One's historical material bears on it more than the model does: the Washingtonians had a written analogue of this Tradition in print within two years of founding. *Algebra and simulation.*

What was assumed. An empty governance row and the multiplier shared with Tradition 6, with the same consequence: its membership figure is identical to Tradition 6's by construction.

There is a further assumption worth naming because the model cannot see past it. Outside issues enter this model only as a dilution of singleness of purpose. Nothing represents a group splitting over a disagreement, losing members to a controversy, or attracting people because of a position it took. Those are the things the Tradition appears to be about, and the model contains none of them, so silence here is not a small effect but an absent mechanism.

In plain terms. Staying out of outside controversies supplies nothing directly and protects single purpose. The model cannot resolve its effect. The history is the more interesting evidence here, and it cuts against the easy story: the Washingtonians wrote down their own version of this rule almost immediately, circulated it in a manual and a newspaper, and dissolved anyway. Whatever preserved AA, it was not this rule on its own, because the other movement had it too.

Tradition 11, attraction rather than promotion

Technical. Raw semantic load 2.69, governing four resources, with visible proof of recovery at 0.49 of that raw load. Principal supplier for Step Two. Tradition 11 has two paths. Its attraction path multiplies inflow from members' Step Twelve practice; its governance row affects four resource columns. In the 400-seed paired factorial, pure attraction loss costs 5.42 members [4.52, 6.33], governance loss costs 2.29 [1.21, 3.36], and combined loss costs 5.88 [4.97, 6.79]. The interaction is 1.83 [0.76, 2.90]. None of the three conditions closes a group in these runs, and each leaves about 98 per cent endpoint-viable. Older tables that set the Tradition itself to zero are mixed interventions, not pure attraction tests. *Algebra and simulation.*

What was assumed. Four non-zero entries: visible proof 0.7, recipient opportunity 0.6, identification 0.2 and continuity 0.2, giving it 44 per cent of visible proof. Unlike Tradition 3 it also has a

direct route into arrivals: the model's inflow is an exogenous term plus a term scaling with the attraction adherence multiplied by the twelfth Step practice in the room.

That asymmetry between Traditions 3 and 11 is the most important assumption in Part Two and it is a deliberate one. The model says a group cannot increase how many people arrive by opening its door wider, because the door is already open by Tradition 3, and can increase it by being visibly worth arriving at. Because attraction moves both the governance path and the arrival path together, an ordinary change to this Tradition's adherence is a mixed intervention, and the primer separates the two paths rather than reporting their sum.

In plain terms. Attraction and resource governance are different jobs in this model. Removing attraction makes the thirty-year group smaller without producing closure in these runs; removing governance also matters. Removing both is not the sum of removing each. No finite simulation establishes that a remnant lasts forever, and the outstanding comparison with referral loss must be reported on closure, endpoint viability and size separately.

Tradition 12, anonymity

Technical. Load 2.62, governing five resources, with confidentiality at 0.57 of its load. Principal supplier for Step Five and the Tradition governing confidentiality at the matrix maximum of 1.0. Degrading it alone costs 0.99 members [0.41, 1.58]. It is also one of two Traditions guarding the effective adherence of the group conscience, entering that term symmetrically with Tradition 9. Because the two enter identically there, the entire difference between their measured losses, 0.99 against 0.02, is associated with Tradition 12's own governance row. In the deliberation model it is the third of the three holding maximum influence near one over N. *Algebra, deliberation and simulation.*

What was assumed. Five non-zero entries, one of them decisive: confidentiality 1.0, identification 0.3, and admission, counsel and continuity at 0.1 each. The 1.0 gives anonymity 59 per cent of all confidentiality in the model, one of the larger single shares any Tradition holds of any resource, though Tradition 3's 71 per cent of admission, the group conscience's 69 per cent of counsel and unity's 60 per cent of gentle pressure are larger.

This is the assumption that produces the cleanest result in the book, and it was made independently of the matching assumption in Step Five's row only in the weak sense that the two were written at different times by the same person. Both encode the same belief, that what anonymity does is make disclosure safe. The primer's Step Five entry gives the numerical size of that belief: the match survives 99.5 per cent of magnitude jitter and 17.5 per cent of full magnitude replacement. Read the two entries together, because neither is independent evidence for the other.

In plain terms. Anonymity does two separate jobs and they are easy to run together. The first is ordinary and immediate: it is what makes confidentiality available, and confidentiality is what the fifth Step needs and cannot do without. The second is structural and invisible from inside the room: by keeping anybody from becoming a name, it stops the group's decisions concentrating on one person, which is the condition for those decisions being reliable at all. Tradition 9 shares the second job and not the first, which is exactly why anonymity scores higher in the membership comparison and why that higher score says nothing about the part that matters most.

That two-job reading was arrived at from the model and has since been found in the source. AA's 1953 commentary on this Tradition tells the two as separate lessons learned at different

times: first that a member's name and story had to be confidential, after members repeated each other's stories and trust broke; and later, once national publicity arrived, that anonymity had to be absolute at press, radio, film and television, so that no self-appointed member could present himself as a messiah representing AA. The first is confidentiality. The second is the structural job. Nobody had to reconcile them because the fellowship never treated them as one thing.

Maxwell wrote in 1950 that anonymity had "sheer survival value" and could not say why. This is the why, and the point worth keeping is that it is two whys rather than one. The Washingtonians, for what it is worth, took the opposite position on this deliberately and with an argument, which is what makes the comparison a comparison between two written codes rather than between rules and no rules.

What the Tradition rows say taken together

Technical. The twelve split into two tiers, and the split is a property of the governance matrix rather than a reading of the text. Seven Traditions govern at least one resource some Step consumes. Five, namely 4, 6, 7, 9 and 10, govern none, so they supply nothing to any Step and appear in the simulation only as multipliers on the adherence of the Traditions they guard. That multiplier form is an assumption. The index-pairing conjecture fails on all twelve counts, and five of the twelve are arithmetic: those five Traditions have empty rows, so no sparsity-preserving perturbation can move the index-mate entry off zero. At every jitter level the proportion of draws in which pairing fails on all twelve equals the proportion in which it fails on the seven that could have gone either way, to the last draw. A threshold test that can reach the five finds them failing by margins comparable to the seven, with thresholds from 0.417 to 0.627 against a mean live governance entry of 0.374. Pairing fails on all twelve in 85.5 per cent [83.9, 87.0] of draws at plus or minus thirty per cent jitter and 40.6 per cent [38.5, 42.8] when every magnitude is randomised.

In plain terms. The author-coded governance table places the Traditions in two kinds; the model did not discover the division independently. Seven of them hand the group something it needs. Five hand over nothing and instead stop something from going wrong: they are guards rather than suppliers. That is a transparent property of one person's coding and an invitation for independent readers to disagree, not a result computation can validate.

The pairing idea, that the first Step goes with the first Tradition and so on down, is wrong everywhere. But the honest version of that result is more careful than the headline. Five of the twelve failures are wrong for a boring reason: those Traditions supply nothing to anybody, so of course they do not supply their own Step. Those five are arithmetic, not evidence, and a separate test had to be built to say anything real about them. It was built, and they fail like the others. What the whole result rests on is whether the strengths in the two tables are roughly right. If they are, the pairing is dead. If a reader accepts only the pattern of which Tradition touches what and rejects every magnitude, it becomes a coin flip. So Part Four argues for its numbers rather than hiding behind a robustness percentage, and this is the one part of the book that works that way.

What this primer does not say

It does not rank the Traditions by importance. The degradation column measures membership at twenty years in a one-factor sweep from 0.85 to 0.50. Seven of the twelve contrasts resolve and five do not. Traditions 2, 9 and 12 carry the book's central argument, and three of the five unresolved rows are protective Traditions whose simulation role is an assumption rather than a derivation.

It does not establish that Traditions 2, 9 and 12 are what prevents the three obstructions. The appendix shows the three obstructions behave as the theorem says. The step from there to the claim that these three Traditions are what prevents them is a reading of three sentences, and it is the book's central claim. Nobody has a method for testing it. In plainer terms: the mathematics says what a group has to avoid, and the Traditions look very much like instructions for avoiding exactly those things, but "looks very much like" is a judgement and no computation upgrades it.

It does not validate the two matrices. Both were built by one person. The eight resources are that person's list and no source proposes it. A second reader marking the ninety-six governance cells is the largest outstanding item in the project and no further computation substitutes for it. Flipping four of fifty-six enabling cells at random leaves the rejection of index-pairing on all twelve counts standing in 86.3 per cent of draws; flipping sixteen leaves it at 51.7, close to a coin flip. In plainer terms: one disagreement here and there is survivable, wholesale disagreement is not, and only a second reader can say which this would be.

Nothing here is calibrated to AA data, because none exists at the required resolution. Inflow, dropout and churn were originally set to target about forty-five members with an experienced core near nine. After mean-centring the lognormal capability draw, 400 runs deliver 17.80 plus or minus 0.88 members overall; among the 394 viable endpoints the experienced count above 0.5 is 1.25 plus or minus 0.20. The calibration fails. It is disclosed rather than repaired after seeing the result.

A follow-up: asking the same questions about a different fellowship

Everything above is about one fellowship. The model was built around AA's twelve Steps and twelve Traditions, and it takes those two lists as its two inputs. A fair question is whether any of it reaches further. In August 2026 that question was asked directly, about Recovery Dharma, a peer-led Buddhist recovery fellowship whose program book is published free under a Creative Commons licence.

The specific question was whether the model shows the twelve Steps simplified into the Dharma. It does not, and the reasons are worth stating here because they mark the edge of what this primer covers.

The dates run the wrong way. The Eightfold Path is roughly two and a half thousand years older than the twelve Steps. Nothing can be a simplification of a document written long after it. Recovery Dharma presents its program as an application of early Buddhist teaching, not as a rewriting of AA.

Counting does not support the word either. Recovery Dharma sets out thirty-five enumerated items across seven lists: three jewels, four Noble Truths, eight path factors, five precepts, four heart practices, four foundations of mindfulness, and seven commitments in a section called

The Practice. AA, as this model codes it, is twenty-four items across two lists. The comparison that makes simplification look true sets the eight path factors against the twelve Steps. That is one list against one list.

There is a group conscience, and it is the sangha. A first pass at this said there was no Traditions equivalent at all. That was wrong, and the correction is worth stating because it turned into the best finding here. Every Recovery Dharma meeting opens with members affirming that they trust in the wisdom of the Buddha, the Dharma and the Sangha, where the Sangha is the community itself. AA locates ultimate authority in a group conscience. Recovery Dharma locates trust in a sangha. That is the same move.

What is missing is not the authority but the procedure. There is no charter setting out numbered provisions. The fellowship's whole governance commitment is one undivided act of trust, rather than twelve separate rules a group could keep or drop one at a time. Searching the whole book turns up no "group conscience", no "consensus", no "business meeting", no "rotation" and no "bylaws". The jobs the Traditions do are real, but several of them sit inside a suggested meeting script that each meeting is invited to edit. Confidentiality sits in the closing. Self-support is the basket passed near the end. The facilitator says plainly that they hold no authority.

And the fellowship exists because of the failure this book is about. Recovery Dharma split in 2019 from an earlier Buddhist recovery program that had been built around a single named teacher. That program fractured, in the words of one of its own former officers, over inequities among its leaders, and people were harmed. The people who rebuilt it, including the predecessor's executive director, made the first commitment of the new fellowship the one every meeting now reads aloud: it is peer-led and follows no one leader or teacher. Nobody involved had heard of this model or of the mathematics behind it. They arrived at Traditions Two and Nine on their own, eighty-four years after AA, by watching what happened without them. It is one case and it is the fellowship's own account of itself, so it corroborates and cannot confirm. Chapter Twenty-Four is where the book uses it and says so.

The deepest difference is the sequence, and it is the one the model can see clearly. Two of the model's central ideas depend on the Steps being worked in order. Growth on each Step is gated by the Step before it. That gate is why the practice which brings in new members, Step Twelve, is also the most expensive one to reach, and why a group's ability to grow lags its ability to hold people. Recovery Dharma's path is grouped under three headings and practiced at the same time rather than in order. Take away the sequence and there is no longer any reason for the member-attracting practice to be the costly one. So those two ideas simply do not carry across.

None of this changes a single figure above. What it changes is the scope a reader should give them. This model is not about recovery groups in general. It is about groups that have a numbered sequence of practices and a fixed written charter. Recovery Dharma has neither, and a model with one Step matrix and one Tradition matrix has nowhere to put a fellowship built that way.

Two cautions, in the spirit of the section above. Nothing here compares the two fellowships for effectiveness, and nothing in this project could. Nothing here is advice about which room anyone should walk into. The full comparison, with the page references and the arithmetic, is appendix section A12, and it is written to stand alone.

A second reading, from the other end. In its founding years the model has most people arrive because a member brought them, rather than finding the group on their own, though by the

thirty-year endpoint the arrivals that do not pass through a member usually supply most of them. That rests on two values the book authored, the rate at which people arrive on their own and the rate at which a member brings someone, and it had never been checked against anything. In August 2026 the thirty personal stories in the 1939 first edition of *Alcoholics Anonymous* were read for exactly this. They support the assumption for that fellowship at that time: the recurring shape is a recovered drinker turning up in person, often several of them, and in one case about twenty men visiting a single man in hospital. Only one story describes somebody reached without a visit, by letter and a copy of the book, and the fellowship presents that as an experiment it was unsure of, recording that it had begun to think the book inadequate without personal contact.

Set beside Recovery Dharma, where almost everybody arrived through a website, a free book, a flier, a therapist or a chaplain, the two readings bound the assumption rather than confirm it. How a fellowship recruits is a fact about that fellowship and that period, not about mutual-aid groups in general, and the model should be read as speaking about AA in its founding decades. Appendix section A13 carries the census and its limits, of which the largest is that the stories were selected to persuade and everyone in them recovered.

Where every figure comes from

The five headings below are the book's canonical reference headings, in the order `tools/check_chapter.py` requires of a chapter. The primer is exempt from that rule, because it is an appendix rather than a chapter and the checker skips its structural block entirely. It keeps the convention anyway, and the exemption is the reason a reader should not assume anything enforced it.

The corpus under `research/staged/` is fully incorporated; what remains there is provenance, not evidence. No file is evidence for this primer merely because it exists locally.

Read in full:

Golub, B. and M. O. Jackson (2010). "Naive Learning in Social Networks and the Wisdom of Crowds." *American Economic Journal: Microeconomics* 2(1): 112-149. The vanishing influence condition and the three obstructions. Read at source.

Maxwell, M. A. (1950). "The Washingtonian Movement." *Quarterly Journal of Studies on Alcohol* 11: 410-452. The claim that anonymity has "sheer survival value". Read in full from the project's retyped reproduction, not at journal source. The original project PDF and text are stored in the Maxwell subdirectory of `research/incorporated/`; four demonstrable transcription errors are listed in `research/SOURCES.md`.

Grosh, A. B. comp. (1842). *Washingtonian Pocket Companion.* Second edition. Utica, N.Y.: B. S. Merrell. The Washingtonians' own manual, used here for the written analogues of Traditions 4, 7, 9 and 10 and for their deliberate opposite position on anonymity. Read at source; held git-ignored in `research/incorporated/Grosh_1842/`.

Recovery Dharma Global (2023). *Recovery Dharma.* Second edition. Recovery Dharma Inc. CC BY-NC-SA 4.0. The only source for the follow-up section above. Read in full, in two passes: the front matter, the whole of Section I, the whole of Section II with its fourteen personal recovery stories, and the glossary, meeting format and dedication of merit on 16 August 2026, and the meditations and inquiry questions on 13 September 2026. The stories were read only after a

first pass had skipped them, and skipping them was the reason that pass got the governance question wrong. The 2019 split is reported in the stories by people who held office in the predecessor organization, whose founder is named in that source and is deliberately not named here. Saved in `research/incorporated/RecoveryDharma_2023/`.

Alcoholics Anonymous World Services (1953). *Twelve Steps and Twelve Traditions.* Read in full on 10 August 2026, from the per-chapter files AAWS publishes free; held as a git-ignored reading copy since 13 September 2026 and never committed. The source behind the Tradition 2, 9 and 12 entries, and for the finding that no Tradition chapter refers to the Step of its own number.

Alcoholics Anonymous World Services. *The Twelve Concepts for World Service* (Short Form), SMF-114 (adopted 1962), read at source on 17 August 2026, and *The A.A. Service Manual Combined with Twelve Concepts for World Service* (2024-2026 edition, BM-31), read in part on 13 September 2026: Concepts IV and V in full, Concept XI on rotation, and every passage mentioning rotation. Both held git-ignored. Behind the Tradition 9 entry: Concept IV proportions voting weight to responsibility and holds no rule about the size of a rotating pool.

Alcoholics Anonymous (first edition, 1939, in the 1999 reprint). The thirty personal stories read in full on 17 August 2026, for the census of how newcomers arrived, in the follow-up section above (appendix A13). Held git-ignored.

Dinerstein, M., R. Megalokonomou and C. Yannelis (2022), NBER Working Paper 27925; Cohen, J. P., A. C. Johnston and A. S. Lindner (2023), NBER Working Paper 31120; Edgren, R., D. Baretta and J. Inauen (2025), *Applied Psychology: Health and Well-Being* 17(1): e12612; and Kaskutas, L. A., J. Bond and L. A. Avalos (2009), *Addictive Behaviors* 34(12): 1029-1035. The four papers that bracket the decay rate, read between 13 and 22 September 2026 (the first two apart from appendices, Edgren apart from its supplement, figures and references, and the Kaskutas paper apart from its figures). Held git-ignored. Chapter Twelve gives what each measures and what it does not.

Jensen, K., and M. Abrahams (2019). "Buddha Buzz Weekly: Refuge Recovery Splits." *Tricycle: The Buddhist Review*, 13 July 2019. Read at source on 17 August 2026; held as record only. The independent account of the 2019 split, which corrects where the contrast with the predecessor falls (appendix A12.4).

Cited at a remove:

Nothing. Every source named here was read at source. The primer restates findings from chapters that do cite at a remove, and those removes are recorded in the chapters rather than repeated here.

Internal, and reproducible from this repository:

`model/aa_group_model.py`, matrices S and GOV, the effective-adherence function, and the arrival and dropout expressions.

`model/book-calculations.ipynb`, whose published-figures cell holds labelled blocks: the part2 closed forms for the influence weights and rotation pools behind the deliberation figures, the part4 blocks for the coupling, the principal suppliers, the index-mate ranks and the threshold test, the "load by resource for each tradition" block for the load column, the "ch17 reassignment test" block, the "ch15 service" block for the Step Twelve service results, the part5 blocks for the failure modes and the Tradition 3 dose sweep, and the coupling perturbation tests for the sparsity perturbation; the resource-list test is `research/resource_list.json` with `model/resource_list_test.py`.

`research/tradition_paired.json`, the 400-replication paired degradation runs, produced by `model/tradition_paired.py`. `research/part5.json`, 4,800 runs behind the Tradition 3 sweep and the three failure modes. `research/ch15_service.json`, the three service configurations. `research/core_thresholds.json`, the two membership thresholds and the calibration figures. `research/resource_list.json`, the sixty-four resource-list variants. `research/oat_full.json`, the 944 registered multi-level perturbation points. `research/structural.json`, the four structural variants. `research/decay_ordering.json` and `research/decay_reversal.json`, the 400-seed paired runs on the decay rate.

`appendix/APPENDIX.md`, sections A2 for the effective-adherence specification, A4 for the seed counts and estimands, A7.1 to A7.5 for the perturbation designs and what they cannot reach, A8 for the sparsity pricing, Chapter Eighteen for the threshold test and Chapter Seventeen for the reassignment test, A5.2 for the degradation ranking, A7.3 for the structural variants and A7.6 for the resource-list test. Part Two's sensitivity analysis is the section titled "Part Two and deterministic coupling tests", which is numbered A8.

Manuscript chapters 8, 9, 10, 12, 13, 15, 16, 17, 18, 19 and 20, whose Machinery sections carry the full designs and estimands for every figure quoted above.

What was not read:

The AA literature was said here to gesture at a parallel between the Steps and the Traditions. That was written without having read it. The 1953 commentary, listed above, was read in full, and no Tradition chapter refers to the Step of its own number, while the word "Tradition" does not appear in any Step chapter at all. So the conjecture is refuted as a thing people believe rather than as a thing somebody published, and I cannot say how strongly the rest of the literature gestures.

No study of AA group culture, which is why the behavioural description behind Tradition 3's retention mechanism is illustration rather than evidence.

Any measurement of how fast a practice lapses when nothing renews it. That literature was searched on 21 September 2026, and four papers from it have now been read. Two on adult skill depreciation, read on 13 September, put six per cent a week one to two orders of magnitude above anything they measure. One on the daily course of habit decay, read on 22 September, is an order of magnitude faster, but it measures an unwanted habit being deliberately weakened. One on seven-year trajectories of AA attendance, read the same day, finds attendance falling away over years while abstinence does not follow it down. The rate is therefore bracketed rather than measured, and it is still a choice. It is a consequential one: with it a fifth lower, a 400-seed run finds that a group without referrals ends larger than a group without attraction but still closes more often, and fifteen per cent lower is not enough to turn it. Chapter Twelve gives the detail.

Anything from the other side about the 2019 split. The account above rests chiefly on the successor fellowship's own literature, written by people who left the predecessor. One independent account, *Tricycle*'s, was read and corrected where the contrast falls, but no statement from the predecessor's side was sought. That is a largely one-sided record and the book's use of it inherits the weakness.

The research literature on Buddhist and mindfulness-based recovery programs, which was not searched at all. The follow-up section compares program documents and one fellowship's account of its own history, and knows nothing about either fellowship's outcomes.